**Note:** The LTTng releases starting from version 2.0 onwards are named after Québec's micro-brewery beers!

## System setup

I am using Fedora as an example; however, you can search for similar packages on Ubuntu and proceed. First, update your system and install the *Development Tools* package, which will give you the kernel headers too. Considering that you have configured *sudo*, give the following commands:

```
$ sudo yum update
$ sudo yum group install "Development tools"
```

Check if the kernel headers package is the same version as your kernel:

```
$ rpm -qa | grep kernel-devel
kernel-devel-3.11.10-301.fc20.x86_64

$ uname -r
3.11.10-301.fc20.x86_64
```

## Building the kernel modules

If all seems to be going well, start off with building and installing the LTTng kernel modules for Fedora. If you are on Ubuntu, skip this step as the *lttng-modules* package is already available in the repos.

```
$ wget http://lttng.org/files/lttng-modules/lttng-modules-2.3.4.tar.bz2
$ tar -xvf lttng-modules-2.3.4.tar.bz2
$ cd lttng-modules-2.3.4.tar.bz2
$ KERNELDIR=/usr/src/kernels/$(uname -r) make
$ sudo KERNELDIR=/usr/src/kernels/$(uname -r) make modules_install
$ sudo depmod -a
$ sudo modprobe lttng-tracer
```

You can use *lsmod | grep lttng* to see if the lttng_tracer module is loaded properly.

## Installing LTTng packages

Install the packages for lttng-tools that provide the main components and the tracing client. You would also need lttng-ust for the userspace tracing library and its *devel* package, which will contain the necessary headers and examples for userspace tracepoints. *Babeltrace*, as described before, is a simple command line CTF trace viewer and converter.

```
$ sudo yum install lttng-tools lttng-ust babeltrace lttng-ust-devel
```

## Post-installation

For kernel tracing, we need the LTTng session daemon to be *run as root*, and the LTTng client to be run either by root itself or by the user who should be part of the 'tracing' group.

```
$ sudo groupadd -r tracing
$ sudo usermod -aG tracing suchakra
```

We are almost set! Just reboot your machine and check if the session daemon (lttng-sessiond) has started automatically and if the LTTng kernel modules are in place or not.

```
$ lsmod | grep lttng_tracer
$ sudo service lttng-sessiond status

$ groups
foo tracing
```

## Your first trace

Let's start with some simple experiments by tracing a 1-second sleep at the kernel level. You can first check the available kernel events:

```
$ lttng list -k
Kernel events:
-------------
      writeback_nothread (loglevel: TRACE_EMERG (0)) (type: tracepoint)
      writeback_queue (loglevel: TRACE_EMERG (0)) (type: tracepoint)
      writeback_exec (loglevel: TRACE_EMERG (0)) (type: tracepoint)
      writeback_start (loglevel: TRACE_EMERG (0)) (type: tracepoint)
      writeback_written (loglevel: TRACE_EMERG (0)) (type: tracepoint)
      writeback_wait (loglevel: TRACE_EMERG (0)) (type: tracepoint)
      writeback_pages_written (loglevel: TRACE_EMERG (0)) (type: tracepoint)
.
.
      sched_switch (loglevel: TRACE_EMERG (0)) (type: tracepoint)
.
```

A long list of available kernel events is shown. Let's pick *sched_switch* for our quick experiment and proceed. Make sure that lttng-session is running as the root before proceeding:

```
#create a new tracing session
$ lttng create osfy
Session osfy created.
Traces will be written in /home/suchakra/lttng-traces/osfy-20131227-220359
```